

House Price Prediction. Predict housing prices from features like area, location, number of rooms, and age of property. Students evaluate using RMSE, MAE, and R^2 .

1. Problem Statement and Problem Formulation

House prices depend on several factors such as the area of the property, location, number of rooms, and age of the property. Accurately estimating the price of a house can help buyers, sellers, real-estate companies, and financial institutions make better decisions.

The objective of this assignment is to develop a Machine Learning-based House Price Prediction System that predicts the price of a house using property-related features.

Input Features

- Area in square feet
- Location
- Number of rooms
- Age of property

Target Variable

- House Price

Since the target variable is continuous, this problem is treated as a regression problem.

2. Objective and Expected Outcomes

Objectives

1. To understand the factors affecting house prices.
2. To preprocess housing data.
3. To perform exploratory data analysis.
4. To visualize relationships between property features and price.
5. To develop regression models for house-price prediction.
6. To compare different regression algorithms.
7. To evaluate the models using MAE, RMSE, and R^2 .
8. To select the most suitable model based on performance.

Expected Outcomes

The system should:

- Successfully preprocess the housing dataset.
- Identify relationships between features and house prices.
- Generate house-price predictions.
- Produce graphs for data analysis.
- Calculate MAE, RMSE, and R^2 .
- Identify the best-performing regression model.

3. Requirements, Constraints and Assumptions

Software Requirements

- Python 3.x
- Google Colab / Jupyter Notebook / VS Code

Python Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor
from sklearn.tree import DecisionTreeRegressor
from sklearn.metrics import mean_absolute_error
from sklearn.metrics import mean_squared_error
from sklearn.metrics import r2_score
```

4. Application of Relevant Course Knowledge

The following concepts are applied:

- Data collection

- Data preprocessing
- Exploratory Data Analysis
- Data visualization
- Categorical data encoding
- Feature selection
- Train-test splitting
- Regression
- Machine Learning
- Model evaluation
- Model comparison

The three primary evaluation metrics are:

MAE

$$MAE = \frac{1}{n} \sum |y_i - \hat{y}_i|$$

MAE represents the average absolute prediction error.

RMSE

$$RMSE = \sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2}$$

RMSE gives greater importance to large prediction errors.

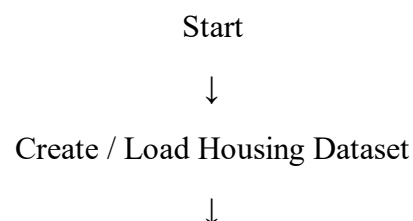
R²

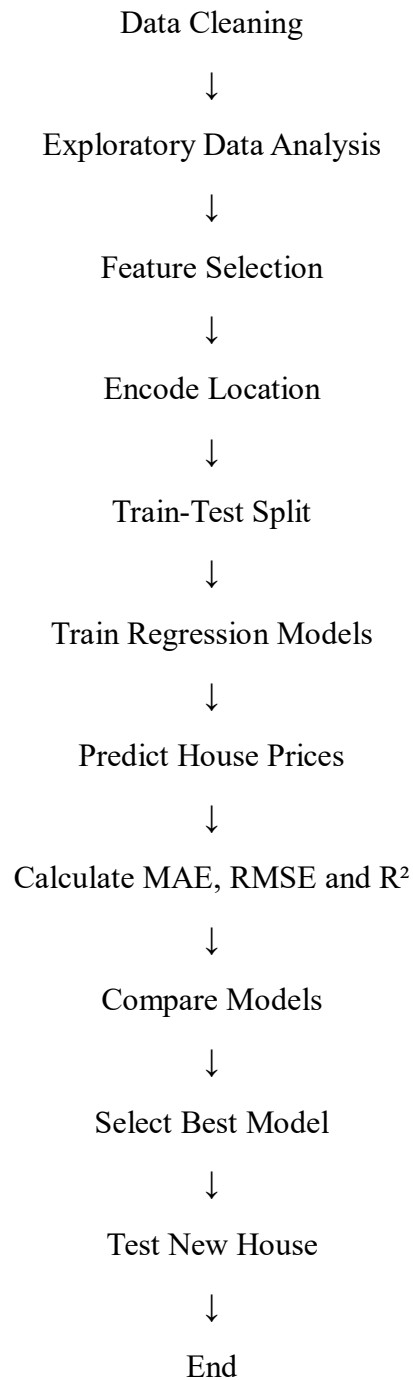
$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

A higher R² indicates that the model explains more variation in house prices.

5. Proposed Solution / Methodology

The proposed system follows these steps:





6. Algorithm

Algorithm: House Price Prediction

1. Start.
2. Create or load the housing dataset.
3. Check the dataset for missing and duplicate values.
4. Remove duplicate records if present.

5. Separate input features and target variable.
6. Encode the categorical location feature.
7. Divide the dataset into training and testing sets.
8. Train regression models.
9. Generate predictions using the test data.
10. Calculate MAE.
11. Calculate RMSE.
12. Calculate R^2 .
13. Compare model performances.
14. Select the best model.
15. Use the selected model to predict the price of a new house.
16. Display the prediction.
17. Stop.

Dataset:

```
data = {  
    "Area": [  
        850, 1200, 1500, 950, 1800, 1100, 1350, 2000, 750, 1600,  
        1250, 1450, 900, 1750, 1050, 2200, 1300, 1550, 800, 1900,  
        1150, 1400, 1700, 1000, 2100, 1250, 1500, 1850, 950, 2300,  
        1350, 1650, 900, 2000, 1100, 1450, 1750, 850, 1950, 1200,  
        1550, 2150, 1000, 1800, 1300, 1600, 750, 2050, 1150, 1400  
    ],  
    "Location": [  
        "Chennai", "Bangalore", "Mumbai", "Chennai", "Bangalore",  
        "Hyderabad", "Chennai", "Mumbai", "Hyderabad", "Bangalore",  
        "Chennai", "Hyderabad", "Mumbai", "Bangalore", "Chennai",  
        "Mumbai", "Hyderabad", "Chennai", "Bangalore", "Mumbai",  
        "Hyderabad", "Chennai", "Bangalore", "Mumbai", "Hyderabad",  
        "Chennai", "Bangalore", "Mumbai", "Hyderabad", "Chennai",  
    ]  
}
```

```

        "Bangalore", "Mumbai", "Hyderabad", "Chennai", "Bangalore",
        "Mumbai", "Hyderabad", "Chennai", "Bangalore", "Mumbai",
        "Hyderabad", "Chennai", "Bangalore", "Mumbai", "Hyderabad",
        "Chennai", "Bangalore", "Mumbai", "Hyderabad", "Chennai"
    ],
    "Rooms": [
        2, 3, 3, 2, 4, 2, 3, 4, 2, 4,
        3, 3, 2, 4, 2, 5, 3, 3, 2, 4,
        3, 3, 4, 2, 5, 3, 3, 4, 2, 5,
        3, 4, 2, 4, 2, 3, 4, 2, 4, 3,
        3, 5, 2, 4, 3, 3, 2, 5, 3, 3
    ],
    "Age": [
        10, 5, 8, 15, 3, 12, 7, 4, 20, 2,
        6, 11, 18, 5, 14, 3, 9, 12, 20, 4,
        10, 7, 2, 16, 5, 8, 6, 3, 17, 2,
        9, 4, 15, 5, 13, 7, 4, 18, 3, 6,
        11, 2, 14, 5, 8, 10, 19, 3, 12, 7
    ],
    "Price": [
        55, 85, 110, 60, 135, 70, 95, 150, 50, 125,
        82, 92, 68, 132, 65, 175, 88, 105, 58, 142,
        78, 98, 128, 62, 165, 84, 108, 138, 57, 180,
        93, 120, 63, 148, 72, 100, 130, 54, 145, 80,
        112, 160, 66, 135, 90, 115, 48, 155, 76, 97
    ]
}

df = pd.DataFrame(data)
print("Dataset created successfully!")
print(df.head())

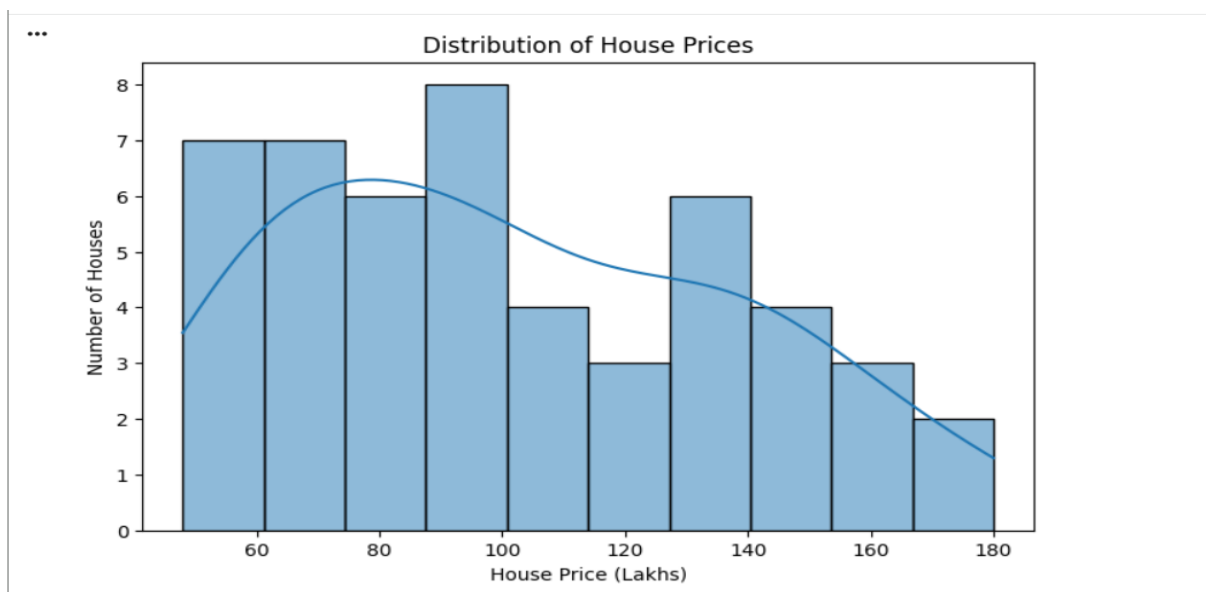
```

10. Exploratory Data Analysis

Graph 1 – House Price Distribution

```
plt.figure(figsize=(8, 5))  
sns.histplot(df["Price"], bins=10, kde=True)  
plt.title("Distribution of House Prices")  
plt.xlabel("House Price (Lakhs)")  
plt.ylabel("Number of Houses")  
plt.show()
```

Output:

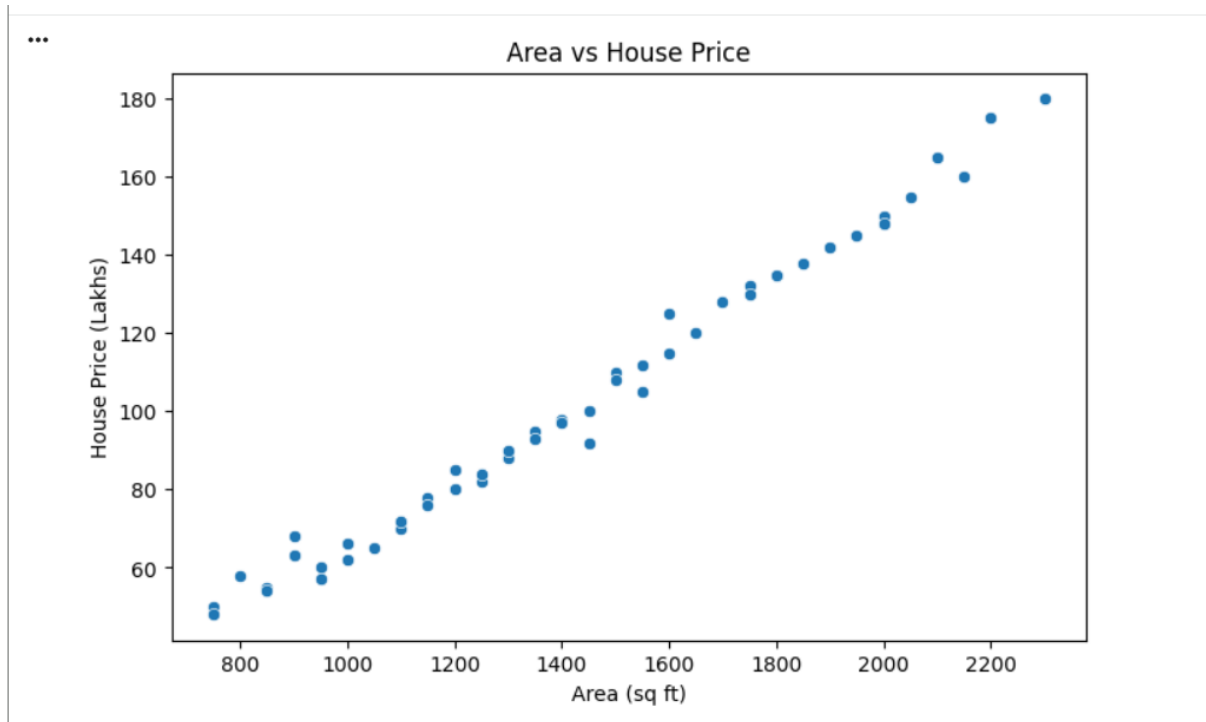


11. Area vs House Price

```
plt.figure(figsize=(8, 5))  
sns.scatterplot(  
    data=df,  
    x="Area",  
    y="Price"  
)  
plt.title("Area vs House Price")  
plt.xlabel("Area (sq ft)")  
plt.ylabel("House Price (Lakhs)")
```

```
plt.show()
```

Output:



12. Rooms vs House Price

```
plt.figure(figsize=(8, 5))
```

```
sns.boxplot(
```

```
    data=df,
```

```
    x="Rooms",
```

```
    y="Price"
```

```
)
```

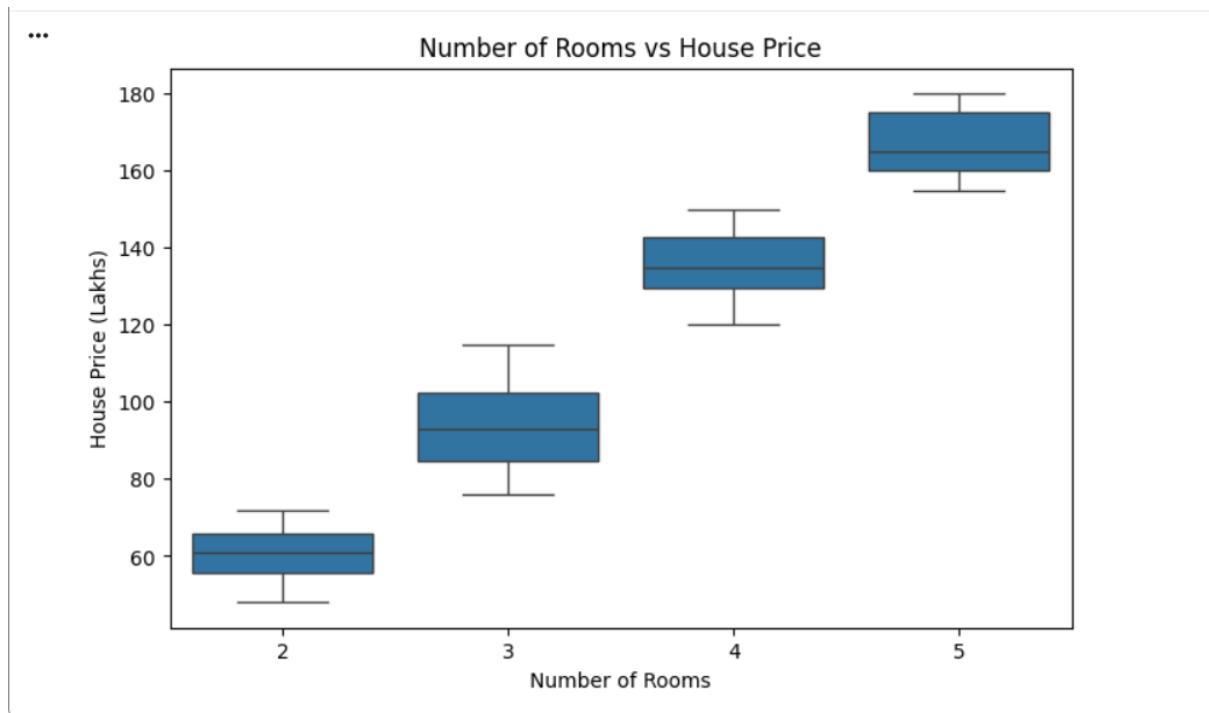
```
plt.title("Number of Rooms vs House Price")
```

```
plt.xlabel("Number of Rooms")
```

```
plt.ylabel("House Price (Lakhs)")
```

```
plt.show()
```


Output:



28. Test Case Table

The report can contain a table in this format:

Test Case	Area	Location	Rooms	Age	Expected Result
TC01	1000	Chennai	2	10	Price should be predicted
TC02	1500	Bangalore	3	5	Price should be predicted
TC03	2000	Mumbai	4	3	Price should be predicted
TC04	1200	Hyderabad	3	8	Price should be predicted

The actual predicted prices should be copied from the Python output after execution.

29. Results and Validation

After executing the program, the model comparison table will contain:

Model	MAE	RMSE	R ²
Linear Regression	Generated by Python	Generated by Python	Generated by Python
Decision Tree	Generated by Python	Generated by Python	Generated by Python

Model	MAE	RMSE	R ²
Random Forest	Generated by Python	Generated by Python	Generated by Python

Important: Do not manually invent these values. Run the code and enter the actual values produced by your Python execution.

Model Selection

The preferred model should generally be the model that provides:

- Lower MAE
- Lower RMSE
- Higher R²

Therefore, the final model should be selected based on the actual experimental results rather than assuming a particular algorithm will always perform best.

30. Analysis and Comparison

The three regression algorithms are compared based on their prediction performance.

Linear Regression

Advantages:

- Simple to implement.
- Easy to understand.
- Computationally efficient.

Limitation:

- May not capture complex nonlinear relationships.

Decision Tree Regression

Advantages:

- Can model nonlinear relationships.
- Easy to interpret.
- Does not require linear relationships between features and price.

Limitations:

- Can overfit the training data.
- Performance depends on tree parameters.

Random Forest Regression

Advantages:

- Combines multiple decision trees.
- Can model complex relationships.
- Usually provides more stable predictions than a single decision tree.

Limitations:

- More computationally expensive.
- Less interpretable than a simple linear model.

The final model shall be selected based on RMSE, MAE, and R^2 obtained from the experiment.

31. Broader Considerations / SDG Relevance

House-price prediction can support data-driven decision-making in the real-estate sector.

It can potentially help:

- Buyers estimate property values.
- Sellers determine reasonable asking prices.
- Real-estate companies analyze market trends.
- Financial institutions assess property values.
- Urban planners understand housing patterns.

Ethical Considerations

The model should be developed responsibly because inaccurate or biased predictions can negatively affect financial decisions.

The dataset should be properly sourced, and sensitive information should not be unnecessarily included.

SDG Relevance

The project can be related to SDG 11 – Sustainable Cities and Communities, particularly through data-driven approaches to understanding housing and urban development.

32. Conclusion

The House Price Prediction system was developed using machine-learning regression techniques. Housing characteristics such as area, location, number of rooms, and property age were used as input features.

The dataset was preprocessed and analyzed using Python. Exploratory data analysis was performed using different graphs to understand relationships between property characteristics and house prices.

Three regression models—Linear Regression, Decision Tree Regression, and Random Forest Regression—were developed and evaluated.

The models were compared using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R^2 Score. The final model should be selected based on the model that provides the most suitable combination of low prediction errors and high R^2 .

33. Limitations

The project has the following limitations:

1. The dataset is relatively small.
2. House prices can depend on many factors that are not included.
3. Location information is represented only at the city level.
4. Real-world housing prices can change over time.
5. The model may not accurately predict unusual or luxury properties.
6. External factors such as economic conditions and market demand are not considered.

34. Possible Improvements

Future versions can be improved by:

- **Using a larger real-world dataset.**
- **Including more locations.**
- **Including number of bathrooms.**
- **Including parking availability.**
- **Including floor number.**
- **Including property type.**
- **Including distance from schools and hospitals.**
- **Including proximity to public transportation.**
- **Applying feature engineering.**
- **Performing hyperparameter tuning.**
- **Using Gradient Boosting or XGBoost.**
- **Deploying the trained model as a web application.**

35. Individual Contribution of Group Members

Team Member Contribution

Member 1	Dataset collection and preprocessing
Member 2	Exploratory data analysis and visualization

Replace the member names and contributions according to your actual team work.

36. References

1. Python Documentation
2. Pandas Documentation
3. NumPy Documentation
4. Matplotlib Documentation
5. Scikit-learn Documentation
6. Seaborn Documentation
7. Housing dataset source, if an external dataset is used
8. Any research papers or websites referred to during the assignment
9. AI-assisted tools used during development, where applicable